

Low-dimensional Laguerre extremality for mixed characteristic polynomials

A global theorem in dimensions two and three, plus balanced local stability in every dimension

Status. Candidate partial result; human review requested.

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Source target. Marcus–Srivastava, *The Solution of the Kadison–Singer Problem*, arXiv:1712.08874. The lecture notes leave an associated-Laguerre extremality assertion unproved:

is at most $(1 + \sqrt{\epsilon})^2$. □

It is worth noting here that the bound $(1 + \sqrt{\epsilon})^2$ is asymptotically tight, as can be seen by picking random vectors so that

$$\mathbb{E} v_i v_i^* = \frac{1}{n} I$$

for all i . The resulting polynomial is an associated Laguerre polynomial whose largest root is (asymptotically) exactly this bound. We believe this polynomial is actually the extremal polynomial for this problem, but are unable to prove it.

The precise conjecture appears in Marcus–Spielman–Srivastava, *Interlacing Families II*. For positive semidefinite $d \times d$ matrices satisfying

$$\sum_{i=1}^m A_i = I_d, \quad \text{tr } A_i \leq \epsilon,$$

it proposes that the largest zero of

$$\mu[A_1, \dots, A_m](x) = \prod_{i=1}^m (1 - \partial_{z_i}) \det \left(x I_d + \sum_{i=1}^m z_i A_i \right) \Big|_{z=0}$$

is maximized by the *capped scalar packing*: as many matrices as possible equal $(\epsilon/d)I_d$, one residual scalar matrix, and the rest zero.

Results. The conjecture is globally true for $d = 2$ and $d = 3$, for every feasible m, ϵ . Moreover, in every dimension, when $m \geq d$ and $\epsilon = d/m$, the balanced scalar tuple $A_i = I_d/m$ is a strict local maximizer along every nonzero affine feasible direction.

Theorem 1 (Global low-dimensional extremality). *For $d \in \{2, 3\}$, the capped scalar packing maximizes the largest zero of $\mu[A_1, \dots, A_m]$ among all feasible positive semidefinite tuples.*

Trace packing. Put $t_i = \text{tr } A_i$. The vector

$$p = (\underbrace{\epsilon, \dots, \epsilon}_{q \text{ entries}}, r, 0, \dots, 0), \quad q = \lfloor d/\epsilon \rfloor, \quad r = d - q\epsilon,$$

majorizes every vector with $0 \leq t_i \leq \epsilon$ and $\sum_i t_i = d$. Hence it simultaneously maximizes every convex power sum $\sum_i t_i^s$, $s \geq 1$.

Dimension two. The determinant expansion and $\sum_i A_i = I_2$ give

$$\mu[A_1, \dots, A_m](x) = x^2 - 2x + 1 - \sum_i \det A_i.$$

Its largest zero is $1 + \sqrt{\sum_i \det A_i}$. If $A \succeq 0$ has trace t , then

$$\det A \leq t^2/4,$$

with equality precisely at $A = (t/2)I_2$. The trace-packing observation now proves the claim.

Dimension three. Let $e_k(A)$ be the k -th elementary symmetric function of the eigenvalues and set

$$S_2 = \sum_i e_2(A_i), \quad S_3 = \sum_i \det A_i.$$

The quadratic coefficient is $3 - S_2$. To find the constant coefficient, expand $\det(\sum_i A_i) = 1$. For each fixed i , the terms with label i twice and another label once sum to

$$\text{tr}(\text{adj}(A_i)(I - A_i)) = e_2(A_i) - 3 \det A_i.$$

Together with the all- i term $\det A_i$, all repeated-label terms sum to $S_2 - 2S_3$. Thus the distinct-label mixed discriminant is $1 - S_2 + 2S_3$, and the exact shifted identity is

$$\mu[A_1, \dots, A_m](1 + y) = y^3 - S_2 y - 2S_3. \quad (1)$$

For $A \succeq 0$ of trace t , Maclaurin's inequalities give

$$e_2(A) \leq t^2/3, \quad \det A \leq t^3/27,$$

with simultaneous equality exactly at $A = (t/3)I_3$. The capped trace packing therefore simultaneously maximizes S_2 and S_3 . Finally, if $a, b \geq 0$ increase in $y^3 - ay - 2b$, evaluating the new polynomial at the old nonnegative largest zero gives a nonpositive value. Its new largest zero cannot be smaller. This proves Theorem 1.

Theorem 2 (Balanced local stability). *Let $m \geq d \geq 2$, $c = 1/m$, and*

$$A_i(t) = cI_d + tH_i, \quad H_i = H_i^*, \quad \text{tr } H_i = 0, \quad \sum_i H_i = 0.$$

For sufficiently small t , let $\lambda(t)$ be the largest zero of the corresponding mixed characteristic polynomial. If some $H_i \neq 0$, then

$$\lambda(t) = \lambda(0) - c_H t^2 + O(t^3), \quad c_H > 0.$$

Proof. Write

$$P_{a,b}(x) = (1 - c \partial_x)^b x^a.$$

For the source's normalized mixed discriminants D , multilinearity and

$$D(H, K) = \text{tr } H \text{ tr } K - \text{tr}(HK)$$

show that every linear perturbation term vanishes. In the coefficient indexed by a k -element subset, the quadratic terms equal

$$c^{k-2}(d-2)_{k-2} \sum_{i < j} -\text{tr}(H_i H_j).$$

Each pair occurs in $\binom{m-2}{k-2}$ subsets, while

$$\sum_{i < j} \text{tr}(H_i H_j) = -\frac{1}{2} \sum_i \text{tr}(H_i^2).$$

Consequently, coefficient by coefficient,

$$\mu_t(x) = P_{d,m}(x) + \frac{t^2}{2} \left(\sum_i \text{tr}(H_i^2) \right) P_{d-2,m-2}(x) + O(t^3). \quad (2)$$

The two P 's are monic scaled associated Laguerre polynomials of degrees d and $d-2$, with the same parameter $m-d \geq 0$. Their zeros strictly interlace. If λ_0 is the largest zero of $P_{d,m}$, then

$$P'_{d,m}(\lambda_0) > 0, \quad P_{d-2,m-2}(\lambda_0) > 0.$$

The largest zero is simple, so the implicit-function theorem and (2) yield

$$c_H = \frac{(\sum_i \text{tr}(H_i^2)) P_{d-2,m-2}(\lambda_0)}{2P'_{d,m}(\lambda_0)} > 0.$$

□

Why this does not yet prove the conjecture in $d = 4$. In dimension four, the same shift has the form

$$\mu[A_1, \dots, A_m](1+y) = y^4 - S_2 y^2 - 2S_3 y + C,$$

where C contains both $-3 \sum_i \det A_i$ and positive $(2, 2)$ mixed-discriminant terms coupling distinct matrices. The capped scalar tuple maximizes S_2, S_3 , but it need not minimize C : the verifier exhibits negative scalar-comparison gaps for this coefficient. Thus the global cubic argument cannot be extended coefficientwise. A proof in $d \geq 4$ needs a new coupled inequality evaluated at the candidate Laguerre root.

Upgrade audit. Five focused routes were completed: exact source/duplicate triage; a direct determinant evaluator and low-dimensional counterexample search; the global $d = 2, 3$ invariant proof; the failed but diagnostic $d = 4$ coefficient comparison; and the all-dimensional balanced second variation. The strongest reproducible run checked 2,700 normalized commuting, rank-one, and full-rank tuples in dimensions three and four with no root violation. This computation is corroborative only.

Scope. The global conjecture remains open in dimension four and higher. Bounded run index and external exact-phrase/keyword searches found the 2015 conjecture and adjacent work, but no source stating the two low-dimensional cases or the balanced Hessian formula. Novelty is plausible but requires specialist review because the $d \leq 3$ proof becomes elementary after the correct shift.

$$\mu[A_1, \dots, A_m](x) = \sum_{k=0}^m x^k (-1)^{m-k} \sum_{S \in \binom{[m]}{k}} \nu(\{A_i\}_{i \in S}).$$

We conjecture that among the families of matrices $A_1, \dots, A_m$ with $\sum_i A_i = I$ and $\text{Tr}(A_i) \leq \varepsilon$, the largest root of the mixed characteristic polynomial is maximized when as many of the matrices as possible equal $\varepsilon I/d$, another is a smaller multiple of the identity, and the rest are zero. When all of the matrices have the same trace, d/m , this produces a scaled associated Laguerre polynomial $L_d^{m-d}(mx)$. The bound that we prove on the largest root of the mixed characteristic polynomial agrees asymptotically with the largest root of $L_d^{m-d}(mx)$ as d/m is held constant and d grows. Evidence for our conjecture may be found in the work of Gurvits [31], [32], who proves that when $m = d$, the constant term of the mixed polynomial is minimized when each A_i equals I/d .

Two natural questions arise from our work. The first is whether one can design an efficient algorithm to find the partitions and pavings that are

Primary sources. A. Marcus and N. Srivastava, *The Solution of the Kadison–Singer Problem*, arXiv:1712.08874. A. Marcus, D. Spielman, and N. Srivastava, *Interlacing Families II: Mixed Characteristic Polynomials and the Kadison–Singer Problem*, Ann. of Math. 182 (2015), 327–350.